

Week 6: Phasors and AC Circuit Analysis

Phasors, impedance, admittance, phasor diagrams, AC nodal, mesh and Thévenin, mixed frequencies

In-depth notes covering all core material of this chapter: theory, engineering shortcuts, exam traps, 9 worked examples, and 8 practice problems with full solutions.

Based on: Rizzoni & Kearns, *Principles and Applications of Electrical Engineering*, 7th ed., §3.4–3.6

Inspired by: ELEC 275, Principles of Electrical Engineering, Concordia University

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How to use this deck

- **“Try it” slides:** the problem alone, with empty space. Do it first.
- **Next slide:** the clean solution, step by step.
- **Purple boxes:** engineering shortcuts, the fast way to see or solve things.
- **Orange boxes:** traps. This is where marks get lost on exams.
- Formulas are tagged **[on formula sheet]** or **[NOT on formula sheet]**, based on the sheet given in ELEC 275 exams.
- **End of deck:** final-check problems, then a full answer sheet.

The right side of each slide is left empty for your notes.

Why phasors?

The problem

With sinusoidal sources, KVL and KCL give differential equations. Solving them with trig identities every time is slow and error-prone.

The fix

In steady state, every signal is a sinusoid at the **same** ω . Only amplitude and phase differ. A phasor stores exactly those two numbers as one complex number, and turns calculus into algebra:

$$\frac{d}{dt} \longrightarrow \times j\omega$$

Result: every week 1–3 method works unchanged, just with complex numbers.

Symbols for this week

Symbol	Unit	Meaning
$\mathbf{V} = V_m \angle \phi$	V	phasor (bold or with a bar): amplitude and phase
$\mathbf{Z} = R + jX$	Ω	impedance: the AC version of resistance
X	Ω	reactance: the imaginary part of $\mathbf{Z}$
$\mathbf{Y} = 1/\mathbf{Z} = G + jB$	S	admittance: the AC version of conductance
G, B	S	conductance, susceptance
ω	rad/s	the one frequency everything runs at

In this course phasors are **cosine-based** and use the **peak** value unless a problem says RMS.

Phasors: the transform

[NOT on formula sheet: memorize]

Time domain

Phasor

$$V_m \cos(\omega t + \phi) \longleftrightarrow$$

$$V_m \angle \phi$$

$$V_m \sin(\omega t + \phi) \longleftrightarrow$$

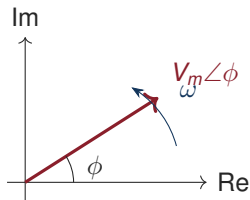
$$V_m \angle (\phi - 90^\circ)$$

$$dv/dt \longleftrightarrow$$

$$j\omega \mathbf{V}$$

$$\int v dt \longleftrightarrow$$

$$\mathbf{V}/(j\omega)$$



the phasor is a snapshot of a vector rotating at ω

Trap: the phasor drops ω . Remember it and put it back when returning to the time domain.

Try it: Example 1, transforms

(a) $v(t) = 100 \cos(1000t + 30^\circ) \text{ V} \rightarrow$ phasor?

(b) $i(t) = -4 \sin(200t + 30^\circ) \text{ A} \rightarrow$ phasor?

(c) $\mathbf{I} = 5 \angle -45^\circ \text{ A}$ at $\omega = 500 \text{ rad/s} \rightarrow i(t)$?

(d) $\mathbf{V} = 6 + j8 \text{ V}$ at $\omega = 50 \text{ rad/s} \rightarrow v(t)$?

your work here ↓

Example 1: solution

(a) already a cosine: $100\angle 30^\circ \text{ V}$

(b) $-\sin(x) = \cos(x + 90^\circ)$: $4 \cos(200t + 120^\circ) \Rightarrow 4\angle 120^\circ \text{ A}$

(c) $i(t) = 5 \cos(500t - 45^\circ) \text{ A}$

(d) polar first: $6 + j8 = 10\angle 53.13^\circ \Rightarrow$

$$v(t) = 10 \cos(50t + 53.13^\circ) \text{ V}$$

Impedance of R, L and C

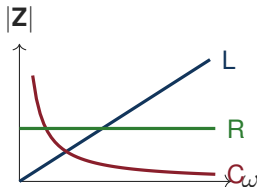
[on formula sheet]

$$R \quad \mathbf{Z}_R = R \quad R \angle 0^\circ$$

$$L \quad \mathbf{Z}_L = j\omega L \quad \omega L \angle 90^\circ$$

$$C \quad \mathbf{Z}_C = -j/(\omega C) \quad \frac{1}{\omega C} \angle -90^\circ$$

$$\mathbf{V} = \mathbf{I}\mathbf{Z} \quad (\text{Ohm's law for phasors})$$



Engineering shortcut

Limits: at DC ($\omega = 0$) L is a wire and C is a gap (week 4 again). At very high ω it flips: L is a gap, C is a wire.

Where impedance comes from

Take $i(t) = I_m \cos(\omega t)$, which is the phasor $\mathbf{I} = I_m \angle 0^\circ$.

Inductor: $v = L \frac{di}{dt} = -\omega L I_m \sin(\omega t) = \omega L I_m \cos(\omega t + 90^\circ)$.

As a phasor: $\mathbf{V} = \omega L I_m \angle 90^\circ = j\omega L \mathbf{I}$, so $\mathbf{Z}_L = \mathbf{V}/\mathbf{I} = j\omega L$.

Capacitor: the same steps with $i = C dv/dt$ give $\mathbf{I} = j\omega C \mathbf{V}$, so

$$\mathbf{Z}_C = \frac{1}{j\omega C} = \frac{-j}{\omega C}.$$

The one idea

Differentiating a sinusoid multiplies its phasor by $j\omega$: the size by ω , and the angle by $+90^\circ$ (since $j = 1 \angle 90^\circ$). That's all "impedance" is: Ohm's law with the derivative already done.

Reactance and admittance

[NOT on formula sheet: memorize]

Impedance: $\mathbf{Z} = R + jX$. $X > 0$ inductive, $X < 0$ capacitive. $X_L = \omega L$, $X_C = -1/(\omega C)$.

Admittance: $\mathbf{Y} = 1/\mathbf{Z} = G + jB$ (siemens). Parallel admittances **add**.

$$\mathbf{Y}_R = 1/R, \quad \mathbf{Y}_L = -j/(\omega L), \quad \mathbf{Y}_C = j\omega C$$

Trap

$G \neq 1/R$ in general. For $\mathbf{Z} = R + jX$: $G = R/(R^2 + X^2)$, $B = -X/(R^2 + X^2)$. Only a pure resistor has $G = 1/R$.

Where G and B come from

Admittance is just $1/\mathbf{Z}$. To split it into real and imaginary parts, multiply by the conjugate:

$$\mathbf{Y} = \frac{1}{R + jX} = \frac{R - jX}{(R + jX)(R - jX)} = \frac{R}{R^2 + X^2} + j \frac{-X}{R^2 + X^2}$$

$$\text{So } G = \frac{R}{R^2 + X^2} \text{ and } B = \frac{-X}{R^2 + X^2}.$$

Only when $X = 0$ does this reduce to $G = 1/R$. With reactance present, the resistor's share of the admittance depends on X too.

When to use admittance

For several branches in **parallel**: add the $\mathbf{Y}$'s (no product-over-sum), then take $1/\mathbf{Y}$ at the end. For series, stay with $\mathbf{Z}$.

Try it: Example 2, impedances

At $\omega = 1000$ rad/s, find the impedance of:

(a) $R = 10 \Omega$ (b) $L = 20$ mH (c) $C = 50 \mu\text{F}$

(d) all three in series: total $\mathbf{Z}$, and is it inductive or capacitive?

(e) the admittance of the $50 \mu\text{F}$ capacitor

your work here ↓

Example 2: solution

(a) $10\ \Omega$

(b) $j(1000)(0.02) = j20\ \Omega$

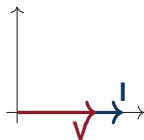
(c) $\frac{-j}{(1000)(50 \times 10^{-6})} = -j20\ \Omega$

(d) $10 + j20 - j20 = 10\ \Omega$: purely resistive (L and C cancel)

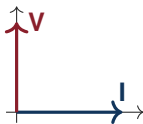
(e) $j\omega C = j0.05\ \text{S}$

(d) is **resonance**: $X_L = |X_C|$, at $\omega_0 = 1/\sqrt{LC}$ (the same ω_0 as week 4).
The circuit looks purely resistive.

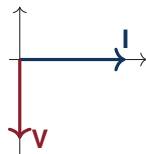
Phase relationships: ELI the ICE man



R: in phase



L: V leads I by 90°



C: I leads V by 90°

Engineering shortcut: “ELI the ICE man”

ELI: in an inductor (L), voltage (E) comes before current (I).

ICE: in a capacitor (C), current (I) comes before voltage (E).

A whole circuit: if **I** lags **V** it's inductive; if **I** leads **V** it's capacitive.

Series, parallel and dividers

[on formula sheet]

Series $\mathbf{Z}_{eq} = \mathbf{Z}_1 + \mathbf{Z}_2 + \dots$

Parallel (two) $\mathbf{Z}_{eq} = \frac{\mathbf{Z}_1 \mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2}$, or add admittances

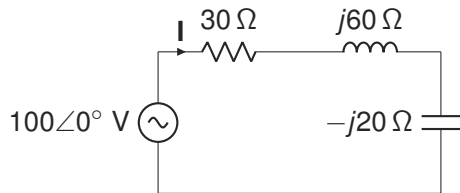
Voltage divider $\mathbf{V}_2 = \mathbf{V} \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2}$

Current divider $\mathbf{I}_1 = \mathbf{I} \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2}$

Trap

Magnitudes don't add. $|\mathbf{V}_R| + |\mathbf{V}_L| + |\mathbf{V}_C|$ can be **bigger** than the source (Example 3). Only the complex sum equals $\mathbf{V}_s$.

Try it: Example 3, series RLC



Find $\mathbf{Z}$, $\mathbf{I}$, and the voltage across each element. Is the circuit inductive or capacitive? Sketch the phasor diagram.

your work here ↓

Example 3: solution (1/2)

Z series impedances add: $30 + j60 - j20 = 30 + j40$; polar:
 $\sqrt{30^2 + 40^2} = 50$, $\tan^{-1}(40/30) = 53.13^\circ$

I Ohm's law $\mathbf{I} = \mathbf{V}/\mathbf{Z}$: divide sizes, subtract angles:

$$\frac{100\angle 0^\circ}{50\angle 53.13^\circ} = 2\angle -53.13^\circ \text{ A}$$

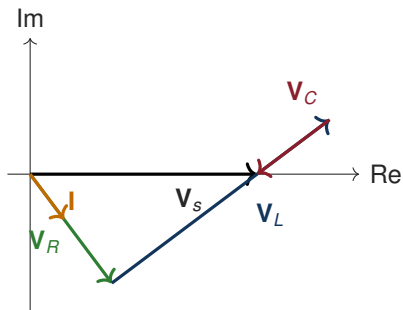
V_R $\mathbf{V} = \mathbf{I}\mathbf{Z}_R$ with $\mathbf{Z}_R = 30\angle 0^\circ$: $60\angle -53.13^\circ \text{ V}$ (in phase with **I**)

V_L $\mathbf{Z}_L = 60\angle 90^\circ$: multiply sizes, add angles: $120\angle 36.87^\circ \text{ V}$
(90° ahead of **I**)

V_C $\mathbf{Z}_C = 20\angle -90^\circ$: $40\angle -143.13^\circ \text{ V}$ (90° behind **I**)

Type net $X = +40 > 0$, and **I** (at -53°) lags **V_s** (at 0°): inductive

Example 3: solution (2/2)

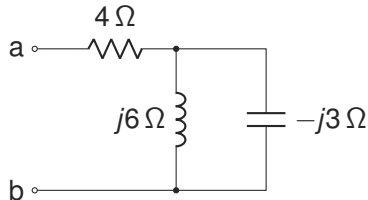


Tip to tail: $\mathbf{V}_R + \mathbf{V}_L + \mathbf{V}_C = \mathbf{V}_s$ (KVL).

$\mathbf{V}_R$ is along $\mathbf{I}$, $\mathbf{V}_L$ is 90° ahead, $\mathbf{V}_C$ is 90° behind.

Note $|\mathbf{V}_L| = 120$ V, bigger than the 100 V source. That's normal.

Try it: Example 4, equivalent impedance



Find $\mathbf{Z}_{ab}$. Is it inductive or capacitive?

your work here ↓

Example 4: solution

Plan combine the parallel L–C pair, then add the series $4\ \Omega$

Top $(j6)(-j3) = -j^2 18 = +18$ (since $j^2 = -1$)

Bottom $j6 + (-j3) = j3$

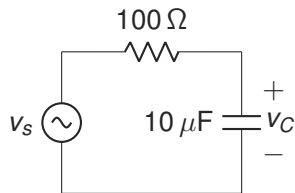
Parallel $\frac{18}{j3} = \frac{6}{j} = -j6\ \Omega$ (dividing by j is multiplying by $-j$)

Series $\mathbf{Z}_{ab} = 4 - j6$; $|\mathbf{Z}| = \sqrt{16 + 36} = 7.21$, angle
 $\tan^{-1}(-6/4) = -56.3^\circ$: $7.21 \angle -56.3^\circ\ \Omega$

Type $X = -6 < 0$: capacitive

Trap: an L in parallel with a C can act capacitive overall. Don't guess from the parts: compute, then read the sign of X .

Try it: Example 5, divider to time domain



$$v_s(t) = 10 \cos(1000t) \text{ V.}$$

Find $v_C(t)$.

your work here ↓

Example 5: solution (1/2)

Plan turn everything into phasors and impedances, use the divider, then return to the time domain

Phasors $v_s = 10 \cos(1000t) \rightarrow 10 \angle 0^\circ$;

$$\mathbf{Z}_C = \frac{-j}{\omega C} = \frac{-j}{(1000)(10^{-5})} = -j100 \, \Omega$$

Divider the C is the “ $\mathbf{Z}_2$ ”: $\mathbf{V}_C = 10 \cdot \frac{-j100}{100 - j100} = 10 \cdot \frac{-j}{1 - j}$
(divide top and bottom by 100)

Example 5: solution (2/2)

Simplify $\frac{-j(1+j)}{(1-j)(1+j)} = \frac{-j+1}{2}$, so $\mathbf{V}_C = 5 - j5$

Polar $\sqrt{25+25} = 7.07$, 4th quadrant: -45°

Time put ω back: $v_C(t) = 7.07 \cos(1000t - 45^\circ) \text{ V}$

AC nodal and mesh: the recipe

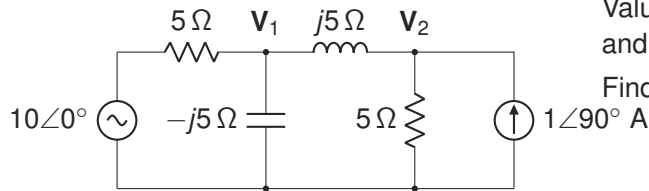
- 1 Convert every source to a phasor (cosine, same ω).
- 2 Convert every R, L, C to its impedance at that ω .
- 3 Solve with nodal or mesh exactly as in week 2, with complex numbers.
- 4 Convert the answers back to the time domain.

Engineering shortcut

In nodal you divide by impedances. Keep $1/j = -j$ handy: $\mathbf{V}/(j5) = -j\mathbf{V}/5$ and $\mathbf{V}/(-j5) = j\mathbf{V}/5$.

Supernodes, supermeshes and dependent sources work exactly as in week 2.

Try it: Example 6, AC nodal



Values are already phasors and impedances.

Find V_1 and V_2 .

your work here ↓

Example 6: solution (1/2)

Plan KCL at both nodes; each current leaving is $(\mathbf{V}_{this} - \mathbf{V}_{other})/\mathbf{Z}$

KCL $\mathbf{V}_1$ leaving through $5\ \Omega$ (to the source), $-j5$ (to ground) and $j5$ (to $\mathbf{V}_2$):
$$\frac{\mathbf{V}_1 - 10}{5} + \frac{\mathbf{V}_1}{-j5} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j5} = 0$$

$\times 5$
$$\frac{5}{-j5} = \frac{1}{-j} = j \text{ and } \frac{5}{j5} = \frac{1}{j} = -j:$$
$$\mathbf{V}_1 - 10 + j\mathbf{V}_1 - j(\mathbf{V}_1 - \mathbf{V}_2) = 0$$

Simplify the $j\mathbf{V}_1$ terms cancel: $\mathbf{V}_1 + j\mathbf{V}_2 = 10$

KCL $\mathbf{V}_2$ leaving through $j5$ and $5\ \Omega =$ source current $1\angle 90^\circ = j$
entering:
$$\frac{\mathbf{V}_2 - \mathbf{V}_1}{j5} + \frac{\mathbf{V}_2}{5} = j$$

Example 6: solution (2/2)

$$\times 5 \quad -j(\mathbf{V}_2 - \mathbf{V}_1) + \mathbf{V}_2 = j5 \Rightarrow j\mathbf{V}_1 + (1 - j)\mathbf{V}_2 = j5$$

Solve $\mathbf{V}_1 = 10 - j\mathbf{V}_2$ into it: $j10 + \mathbf{V}_2 + (1 - j)\mathbf{V}_2 = j5 \Rightarrow (2 - j)\mathbf{V}_2 = -j5$

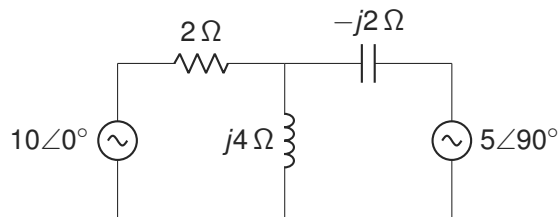
Divide $\mathbf{V}_2 = \frac{-j5(2 + j)}{(2 - j)(2 + j)} = \frac{5 - j10}{5}:$

$$\mathbf{V}_2 = 1 - j2 = 2.24 \angle -63.4^\circ \text{ V}$$

Back $\mathbf{V}_1 = 10 - j(1 - j2) = 10 - j - 2:$

$$\mathbf{V}_1 = 8 - j = 8.06 \angle -7.1^\circ \text{ V}$$

Try it: Example 7, AC mesh



Both sources: + on top. Find the clockwise mesh currents $\mathbf{I}_1$, $\mathbf{I}_2$ and the current in the inductor.

your work here ↓

Example 7: solution (1/2)

Matrix by inspection: diagonal = total impedance around each mesh;
off-diagonal = $-(\text{shared impedance})$; right side = source rises walking
clockwise.

Mesh 1 $2 + j4$ around it, shares $j4$ with mesh 2; the 10 V is a rise going up:
row $[2 + j4, -j4]$, RHS 10

Mesh 2 $j4 - j2 = j2$ around it; walking down the right side enters the $5\angle 90^\circ$
source at its $+$ (a drop): RHS $-j5$

$$\text{det } (2 + j4)(j2) - (-j4)^2 = j4 + j^2 8 - j^2 16 = j4 - 8 + 16 = 8 + j4$$

Example 7: solution (2/2)

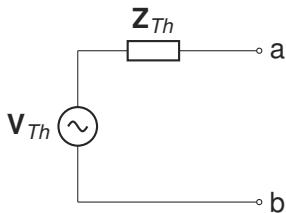
$$\mathbf{I_1} \quad \text{Cramer: } \frac{10(j2) - (-j4)(-j5)}{8 + j4} = \frac{j20 + 20}{8 + j4} \Rightarrow$$

$$\boxed{3 + j = 3.16 \angle 18.4^\circ \text{ A}}$$

$$\mathbf{I_2} \quad \frac{(2 + j4)(-j5) - (-j4)(10)}{8 + j4} = \frac{20 + j30}{8 + j4} \Rightarrow \boxed{3.5 + j2 \text{ A}}$$

Inductor shared branch: $\mathbf{I_1} - \mathbf{I_2} = -0.5 - j$: $\boxed{1.12 \angle -116.6^\circ \text{ A}}$
(down)

AC Thévenin and Norton



Exactly the week 3 methods, with impedances:

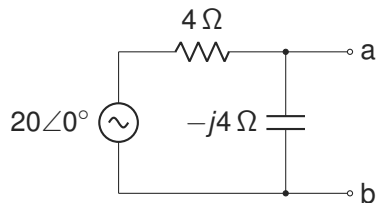
$V_{Th} = V_{oc}$, Z_{Th} with independent sources killed,

$I_N = I_{sc}$, $Z_{Th} = V_{oc}/I_{sc}$.

Trap

Z_{Th} is only valid at **one** frequency. Change ω and every $j\omega L$ and $1/j\omega C$ changes, so the equivalent changes too.

Try it: Example 8, AC Thévenin



Find $\mathbf{V}_{Th}$ and $\mathbf{Z}_{Th}$ at a–b.

your work here ↓

Example 8: solution

V_{Th} open a–b: no load current, so a–b is simply across the capacitor, and the $4\ \Omega$ and $-j4$ form a divider

$$20 \cdot \frac{-j4}{4 - j4} = 20 \cdot \frac{-j}{1 - j} = 20 \cdot \frac{1 - j}{2} = 10 - j10:$$

$$\mathbf{V_{Th} = 14.1 \angle -45^\circ\ V}$$

Z_{Th} kill the source (wire): from a–b the $4\ \Omega$ and $-j4$ are in parallel:

$$\frac{4(-j4)}{4 - j4} = \frac{-j16(4 + j4)}{32} = \mathbf{2 - j2\ \Omega}$$

Meaning at this frequency the network acts like $14.1 \angle -45^\circ\ V$ in series with $2\ \Omega$ and a capacitive $-j2\ \Omega$

Sources at different frequencies

Never add phasors of different frequencies

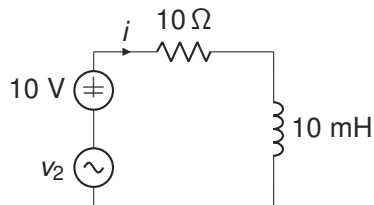
A phasor has no ω in it, so $5\angle 0^\circ$ at 100 rad/s and $3\angle 0^\circ$ at 200 rad/s look alike but can't be combined.

Use superposition

1. Solve the circuit once per source, each at **its own** ω (impedances change with ω).
2. Convert each answer to the time domain.
3. Add in the **time domain**.

A DC source is a source at $\omega = 0$: L is a wire, C is a gap.

Try it: Example 9, DC + AC



A 10 V DC source is in series with
 $v_2 = 10 \cos(1000t)$ V.
Find $i(t)$.

your work here ↓

Example 9: solution

Plan DC ($\omega = 0$) and AC (1000 rad/s) are different frequencies: solve each alone, add in time

DC alone kill the AC source (wire); at DC the inductor is a wire:
 $i_{DC} = 10/10 = 1 \text{ A}$

AC alone kill the DC source (wire); $\mathbf{Z}_L = j(1000)(0.01) = j10$, so
 $\mathbf{Z} = 10 + j10 = 14.14\angle 45^\circ$

Phasor $\mathbf{I} = \frac{10\angle 0^\circ}{14.14\angle 45^\circ} = 0.707\angle -45^\circ \text{ A}$

Add (time) $i(t) = 1 + 0.707 \cos(1000t - 45^\circ) \text{ A}$

Trap: “ $1 + 0.707\angle -45^\circ$ ” mixes a DC value with a phasor, which is meaningless. The sum only exists in the time domain.

Traps

Check these before you box an answer

- Convert sin sources to cos before taking the phasor.
- $\mathbf{Z}_C = -j/(\omega C)$: the minus sign is easy to lose.
- Use ω in rad/s, not f in Hz. $\omega = 2\pi f$.
- Don't add magnitudes. Only complex sums obey KVL and KCL.
- Inductive or capacitive: look at the sign of X after combining, not the parts.
- Different frequencies: separate solutions, then add in time.

Exam-format traps

Give the form they ask for

“Find $v(t)$ ” needs a time function with ω . “Find $\mathbf{V}$ ” wants a phasor. Give polar form unless told otherwise.

Peak or RMS

State whether a phasor magnitude is peak or RMS. Power problems (week 7) usually use RMS.

Calculator

Do complex division in polar or with your calculator’s complex mode. Hand rectangular division under exam pressure is where most errors happen.

Summary (1/2)

Phasor	$V_m \cos(\omega t + \phi) \leftrightarrow V_m \angle \phi; d/dt \rightarrow j\omega$
Impedance	$R, j\omega L, -j/(\omega C); \mathbf{V} = \mathbf{IZ}$
Reactance	$X > 0$ inductive (I lags), $X < 0$ capacitive (I leads)
Admittance	$\mathbf{Y} = 1/\mathbf{Z} = G + jB$; parallel admittances add
ELI the ICE man	L: V leads I by 90° ; C: I leads V by 90°

Summary (2/2)

Combining Analysis	series add, parallel product over sum, dividers as usual phasors $\rightarrow$ impedances $\rightarrow$ week 2–3 methods $\rightarrow$ back to time
Thévenin	$\mathbf{V}_{oc}$, $\mathbf{Z}_{Th}$ at one ω
Mixed ω	superposition, one solution per frequency, add in time
Resonance	$X_L = X_C $ at $\omega_0 = 1/\sqrt{LC}$: purely resistive

Final check

8 problems. One per slide, with space. No looking back.

If one takes more than 10 min, go back to that section. The answer sheet comes after the last problem.

P1: transforms

(a) $i(t) = -4 \sin(200t + 30^\circ)$ A $\rightarrow$ phasor.

(b) $\mathbf{V} = 6 + j8$ V at $\omega = 50$ rad/s $\rightarrow v(t)$.

(c) $v(t) = 12 \cos(377t - 60^\circ)$ V $\rightarrow$ phasor.

your work here $\downarrow$

P2: impedances at 60 Hz

$f = 60 \text{ Hz}$. $R = 20 \Omega$, $L = 100 \text{ mH}$, $C = 100 \mu\text{F}$, all in series.

Find $\mathbf{Z}_L$, $\mathbf{Z}_C$, the total $\mathbf{Z}$, and say whether it's inductive or capacitive.

your work here ↓

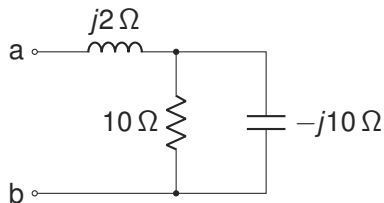
P3: series RL

$\mathbf{V}_s = 50\angle 0^\circ$ V drives $R = 6\ \Omega$ in series with $\mathbf{Z}_L = j8\ \Omega$.

Find $\mathbf{I}$, $\mathbf{V}_R$ and $\mathbf{V}_L$. Check that $\mathbf{V}_R + \mathbf{V}_L = \mathbf{V}_s$.

your work here ↓

P4: equivalent impedance



Find $\mathbf{Z}_{ab}$ in both forms. Inductive or capacitive?

your work here ↓

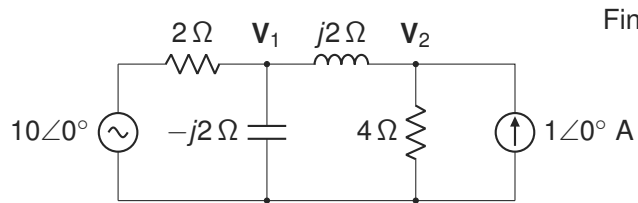
P5: divider in the time domain

$v_s(t) = 20 \cos(500t)$ V drives $R = 40 \, \Omega$ in series with $L = 80$ mH.

Find $v_L(t)$, the voltage across the inductor.

your work here ↓

P6: AC nodal



Find V_1 and V_2 .

your work here ↓

P7: AC Thévenin

A $10\angle 0^\circ$ V source is in series with $5\ \Omega$. Terminals a–b are across an inductor $\mathbf{Z}_L = j5\ \Omega$ connected after the $5\ \Omega$ (like Example 8, with L instead of C).

Find $\mathbf{V}_{Th}$ and $\mathbf{Z}_{Th}$.

your work here ↓

P8: DC + AC

A 12 V DC source and $v_2 = 6 \cos(100t)$ V are in series, driving $R = 4 \Omega$ in series with $L = 30$ mH.

Find $i(t)$.

your work here ↓

Answer sheet

Full worked solutions to P1–P8.

P1: solution

(a) convert phasors need a positive cosine: $-\sin(x) = \cos(x + 90^\circ)$,
so $i = 4 \cos(200t + 120^\circ)$

(a) phasor read amplitude and angle: $4 \angle 120^\circ \text{ A}$

(b) polar $6 + j8$: $r = \sqrt{36 + 64} = 10$, $\theta = \tan^{-1}(8/6) = 53.13^\circ$
(1st quadrant)

(b) time put ω back: $v(t) = 10 \cos(50t + 53.13^\circ) \text{ V}$

(c) already a cosine: $12 \angle -60^\circ \text{ V}$

P2: solution

ω impedances need rad/s: $\omega = 2\pi f = 2\pi(60) = 377 \text{ rad/s}$

$$\mathbf{Z}_L \quad j\omega L = j(377)(0.1) = j37.7 \, \Omega$$

$$\mathbf{Z}_C \quad \frac{-j}{\omega C} = \frac{-j}{(377)(100 \times 10^{-6})} = -j26.5 \, \Omega$$

Total series impedances add: $20 + j37.7 - j26.5 = 20 + j11.2 \, \Omega$
 $= 22.9 \angle 29.2^\circ$

Type net reactance $+11.2 > 0$: inductive (the inductor “wins”)

P3: solution

Plan series circuit: total impedance, then the current, then each element's voltage with $\mathbf{V} = \mathbf{IZ}$

$$\mathbf{Z} \quad 6 + j8: \sqrt{36 + 64} = 10, \tan^{-1}(8/6) = 53.13^\circ: 10\angle 53.13^\circ$$

$$\mathbf{I} \quad 50\angle 0^\circ / 10\angle 53.13^\circ = 5\angle -53.13^\circ \text{ A}$$

$$\mathbf{V}_R \quad 6 \times 5\angle -53.13^\circ = 30\angle -53.13^\circ \text{ V} = 18 - j24$$

$$\mathbf{V}_L \quad 8\angle 90^\circ \times 5\angle -53.13^\circ = 40\angle 36.87^\circ \text{ V} = 32 + j24$$

Check KVL with complex numbers: $(18 - j24) + (32 + j24) = 50 \checkmark$
(but $30 + 40 \neq 50$: sizes don't add)

P4: solution

Plan combine the parallel pair first, then add the series inductor

Parallel product over sum: $\frac{10 \times (-j10)}{10 - j10} = \frac{-j100}{10 - j10}$

Divide multiply top and bottom by the conjugate $10 + j10$:

$$\frac{-j100(10 + j10)}{10^2 + 10^2} = \frac{1000 - j1000}{200} = 5 - j5 \Omega$$

Series add the $j2$: $5 - j5 + j2 = 5 - j3 = 5.83 \angle -31.0^\circ \Omega$

Type $X = -3 < 0$: capacitive

P5: solution

Phasors $\mathbf{V}_s = 20\angle 0^\circ$, $\mathbf{Z}_L = j\omega L = j(500)(0.08) = j40\ \Omega$

Divider $\mathbf{V}_L = \mathbf{V}_s \frac{\mathbf{Z}_L}{R + \mathbf{Z}_L} = 20 \cdot \frac{j40}{40 + j40} = 20 \cdot \frac{j}{1 + j}$

Simplify $\frac{j}{1 + j} \cdot \frac{1 - j}{1 - j} = \frac{j + 1}{2}$, so $\mathbf{V}_L = 10 + j10 = 14.14\angle 45^\circ$

Time put ω back: $v_L(t) = 14.14 \cos(500t + 45^\circ)\text{ V}$

Sense the inductor's voltage leads the source by 45° and is $1/\sqrt{2}$ of it, since $|X_L| = R$

P6: solution (1/2)

Plan same circuit shape as Example 6: KCL at both nodes, multiply out the j 's, substitute

$$\text{KCL } \mathbf{V}_1 \quad \frac{\mathbf{V}_1 - 10}{2} + \frac{\mathbf{V}_1}{-j2} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j2} = 0; \times 2 \text{ with } \frac{1}{-j} = j, \frac{1}{j} = -j:$$
$$\mathbf{V}_1 + j\mathbf{V}_2 = 10$$

$$\text{KCL } \mathbf{V}_2 \quad \frac{\mathbf{V}_2 - \mathbf{V}_1}{j2} + \frac{\mathbf{V}_2}{4} = 1; \times 4: -j2(\mathbf{V}_2 - \mathbf{V}_1) + \mathbf{V}_2 = 4 \Rightarrow$$
$$j2\mathbf{V}_1 + (1 - j2)\mathbf{V}_2 = 4$$

P6: solution (2/2)

Substitute $\mathbf{V}_1 = 10 - j\mathbf{V}_2$: $j20 + 2\mathbf{V}_2 + (1 - j2)\mathbf{V}_2 = 4 \Rightarrow$
 $(3 - j2)\mathbf{V}_2 = 4 - j20$

Divide $\mathbf{V}_2 = \frac{(4 - j20)(3 + j2)}{13} = \frac{52 - j52}{13}$:

$$4 - j4 = 5.66\angle -45^\circ \text{ V}$$

Back $\mathbf{V}_1 = 10 - j(4 - j4) = 6 - j4$: $7.21\angle -33.7^\circ \text{ V}$

P7: solution

Setup source $10\angle 0^\circ$, series $5\ \Omega$, and $\mathbf{Z}_L = j5\ \Omega$ across a–b

$\mathbf{V}_{Th}$ open a–b: divider onto the inductor:

$$10 \cdot \frac{j5}{5 + j5} = 10 \cdot \frac{j}{1 + j} = 5 + j5 = 7.07\angle 45^\circ\text{ V}$$

$\mathbf{Z}_{Th}$ kill the source (wire): $5 \parallel j5 = \frac{j25}{5 + j5} = \frac{j25(5 - j5)}{50} =$

$$2.5 + j2.5\ \Omega$$

Type positive reactance: the equivalent looks **inductive** at this frequency

P8: solution

Plan two different frequencies (DC is $\omega = 0$): solve each alone, then add in the time domain

DC alone inductor $\rightarrow$ wire: $i_{DC} = 12/4 = 3 \text{ A}$

AC alone $\mathbf{Z}_L = j(100)(0.03) = j3$, so $\mathbf{Z} = 4 + j3 = 5\angle 36.87^\circ$

Phasor $\mathbf{I} = \frac{6\angle 0^\circ}{5\angle 36.87^\circ} = 1.2\angle -36.87^\circ \text{ A}$

Add in time

$$i(t) = 3 + 1.2 \cos(100t - 36.87^\circ) \text{ A}$$